Practice runs and evidence

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use repeated trials to prepare an evidence-based design explanation.

SUCCESS CRITERIA - I can...

1. Measure the actual course distance and run times.
2. Calculate average speed and describe variation.
3. Choose a final change supported by repeated evidence.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “trial” and “mean” to explain your model or result. Sentence starters are optional.

trial: one run or repetition of a test

mean: sum of the values divided by the number of values

evidence: observations or measurements that support an idea

01 Bring back an earlier idea

A 50-tooth driver turns a 350-tooth gear.
Give ideal output/input torque and speed ratios and state the loss assumption.

Word help: torque = the turning effect of a force about an axis

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: mean time = sum of trial times/number of trials
average speed = distance/time.
Measure course length; keep release and timing consistent.
Retain every trial and required build evidence.

Worked reasoning: Supplied 3 m runs take 2.0, 2.4, 2.1 s. Report all trials; their mean time is 2.17 s. “Fastest run 2.0 s” answers a different question from typical performance.

Your turn: Make a trial table and mark both mean and fastest time.

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: ① ② ③

03 Represent and interpret

Complete a final labeled sketch or attach a photo with labels.
Identify a feature that supports straight motion and one that supports reliable electrical operation.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

04 Solve and justify

Supplied times over 4.0 m are 5.0, 5.2 and 4.8 s.
Find mean time, time range and speed calculated from mean time.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Analyze the reasoning

One run stops early after a loose connection. Should you erase it?
Explain how to record it fairly and decide whether it belongs in a speed comparison.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3

06 Choose a method

The same flux change through the same coil occurs in 1/4 of the time.
Compare induced-emf magnitude and name a condition for that comparison.

Word help: flux = a measure of magnetic field passing through a chosen area

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: 1 2 3

07 Exit check

Write a three-sentence claim-evidence-reasoning explanation for a design choice. Distinguish a controlled comparison from a plausible mechanism only.

Sentence starter: My design shows ... . It meets the requirement by ... .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

For 3 m runs of 2.0 and 3.0 s, compare mean of the two speeds with total distance divided by total time.
Explain why the results differ.

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: 1 2 3